# H6: Magnetic fields in matter

## 6.1 magnetization

### 6.1.1 diamagnets, paramagnets, ferromagnets

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| magnets microscopic | $e^-$ produce a current around the nuclei of atoms<br>> these are magnetic dipoles |
| magnetization | normally the atomic magnetic dipoles are randomly orientated<br>> cancel each other out<br>> apply a magnetic field B, the domains become aligned<br>> medium becomes magnetically polarized |
| types of magnetization | if we apply a magnetic field B, the we have 3 types of magnetization<br><br>- paramagnets: material magnetizes parallel to B<br>- diamagnets: material magnetizes opposite to B<br>- ferromagnet: material retains magnetization after B is removed<br><br>>> the magnetization isn't only dependent on the current applied field B<br>> it is also dependent on the magnetic 'history' of the object |

### 6.1.2 torques and forces on magnetic dipoles

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| torque on a loop | consider a rectangular current loop in a uniform field B<br>> centre the loop at the origin with tilt θ from the z-axis towards the y-axis<br>> let B point in the z-direction<br>> forces on the two sloping sides cancel<br>on the 'horizontal' sides are equal and opposite<br>> net force is 0, but they create a torque:<br>$\mathbf{N} = aF\sin\theta\,\hat{\mathbf{x}}.$<br>with a force on each segment:<br>$F = IbB,$<br>thus:<br>$\mathbf{N} = IabB\sin\theta\,\hat{\mathbf{x}} = mB\sin\theta\,\hat{\mathbf{x}},$<br>or rather:<br>$\mathbf{N} = \mathbf{m}\times\mathbf{B},$<br>with m = Iab |
| force on a loop | for an infinitesimal loop with dipole moment m in a field B:<br>$\mathbf{F} = \nabla(\mathbf{m}\cdot\mathbf{B})$ |

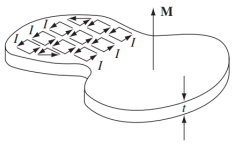
### 6.1.4 magnetization

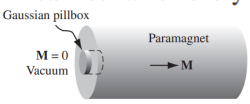
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| magnetization M | = magnetic dipole moment per unit volume |

## 6.2 the field of a magnetized object

### 6.2.1 bound currents

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| potential of magnetized object | consider a piece of magnetized material with magnetization M<br>> the vector potential of a single dipole m is given by:<br>$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\hat{\boldsymbol{\imath}}}{\imath^2}.$<br>thus the total potential is: $\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int\frac{\mathbf{M}(\mathbf{r}')\times\hat{\boldsymbol{\imath}}}{\imath^2}\,d\tau'.$<br>using the identity: $\boldsymbol{\nabla}'\frac{1}{\imath} = \frac{\hat{\boldsymbol{\imath}}}{\imath^2}.$<br>we have: $\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int\left[\mathbf{M}(\mathbf{r}')\times\left(\boldsymbol{\nabla}'\frac{1}{\imath}\right)\right]d\tau'.$<br><=> $\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\left\{\int\frac{1}{\imath}[\boldsymbol{\nabla}'\times\mathbf{M}(\mathbf{r}')]\,d\tau' - \int\boldsymbol{\nabla}'\times\left[\frac{\mathbf{M}(\mathbf{r}')}{\imath}\right]d\tau'\right\}$<br>define the *bound currents*:<br>$\mathbf{J}_b = \boldsymbol{\nabla}\times\mathbf{M},$ and $\mathbf{K}_b = \mathbf{M}\times\hat{\mathbf{n}},$<br>then:<br>$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int_{\mathcal{V}}\frac{\mathbf{J}_b(\mathbf{r}')}{\imath}\,d\tau' + \frac{\mu_0}{4\pi}\oint_{\mathcal{S}}\frac{\mathbf{K}_b(\mathbf{r}')}{\imath}\,da'.$ |

| **6.2.2 physical interpretation of bound currents** | |
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| interpretation of K | consider a magnetized object<br>> the object consists of aligned dipoles<br>> the dipoles in the material cancel each other, since a north and south pole are connected<br>> however, at the edges this doesn't cancel<br>> this is equivalent to a single ribbon of current I flowing around the boundary<br><br>  <br><br>say each tiny loop has area a and thickness t<br>> if we have magnetization M, he dipole moment of each loop is m=Mat<br>> in terms of current, we have m=Ia<br>> therefore: I = Mt, such that the surface current is:<br>$$K_b = I/t = M.$$<br>for generally, we have:<br>$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}.$$<br>we thus have an macroscopic affect of a current looping around the object<br>> however, microscopic, these are bound charges moving in tiny loops |
| interpretation of J | if the magnetization is nonuniform, the internal currents don't cancel<br>> on the surface where two currents join, the net current in the x-direction is:<br>$$I_x = [M_z(y+dy) - M_z(y)]\,dz = \frac{\partial M_z}{\partial y}\,dy\,dz.$$<br>thus, the current density is:<br>$$(J_b)_x = \frac{\partial M_z}{\partial y}.$$<br>a nonuniform magnetization in the y-direction would contribute an amount -∂M/∂z, thus:<br>$$(J_b)_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z}.$$<br>thus, more generally:<br>$$\mathbf{J}_b = \nabla \times \mathbf{M},$$<br>like any other steady current, J should obey the conservation law:<br>$$\nabla \cdot \mathbf{J}_b = 0.$$ |
| **6.3 the auxiliary field H** | |
| **6.3.1 Ampère's law in magnetized materials** | |
| free current | = field due to the bound currents + field due to everything else:<br>$$\mathbf{J} = \mathbf{J}_b + \mathbf{J}_f.$$ |
| auxiliary field H | with the free current, Ampère's law can be written as:<br>$$\frac{1}{\mu_0}(\nabla \times \mathbf{B}) = \mathbf{J} = \mathbf{J}_f + \mathbf{J}_b = \mathbf{J}_f + (\nabla \times \mathbf{M}),$$<br><=><br>$$\nabla \times \left(\frac{1}{\mu_0}\mathbf{B} - \mathbf{M}\right) = \mathbf{J}_f.$$<br>define:<br>$$\mathbf{H} \equiv \frac{1}{\mu_0}\mathbf{B} - \mathbf{M}.$$<br>then:<br>$$\nabla \times \mathbf{H} = \mathbf{J}_f,$$<br>or in integral form:<br>$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{f_{enc}},$$ with $I_{fenc}$ the free current through the Ampèrian loop |
| **6.3.2 a deceptive parallel** | |
| divergence of H | The divergence of H is NOT 0, rather:<br>$$\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}.$$ |

| 6.4 linear and nonlinear media | |
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| **6.4.1 magnetic susceptibility and permeability** | |
| linear midia | in paramagnetic and diamagnetic materials the magnetization is due to field B<br>> if B is removed, M disappears<br>> we have:<br>$\mathbf{M} = \chi_m \mathbf{H}.$<br>with $\chi_m$ the magnetic susceptibility<br>> materials that obey this law are *linear media*<br>> then:<br>$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0(1 + \chi_m)\mathbf{H},$<br>B is proportional to H by:<br>$\mathbf{B} = \mu \mathbf{H},$<br>with μ the permeability:<br>$\mu \equiv \mu_0(1 + \chi_m).$ |
| divergence of M | on the boundary of two different materials the divergence of M can be infinite<br>ex: Gaussian pillbox where the inside has M=0 and outside has an M<br>> therefore: $\oint \mathbf{M} \cdot d\mathbf{a} \neq 0,$ and $\nabla \cdot \mathbf{M}$ cannot vanish everywhere within it.<br>  |
| volume bound and free current density | The volume bound current density in a homogeneous linear material is proportional to the free current density:<br>$\mathbf{J}_b = \nabla \times \mathbf{M} = \nabla \times (\chi_m \mathbf{H}) = \chi_m \mathbf{J}_f.$ |
| **6.4.2 ferromagnetism** | |
| ferromagnetism | = permanent magnetic materials<br>> material consists of permanent magnetic dipoles<br>> the alignment of these dipoles determines the total magnetic moment<br>> these are usually randomly aligned, thus there is usually no magnetic moment<br>> however, a magnet can align these dipoles, creating a permanent magnet<br>> these dipoles keep aligned, due to torque on each other<br>> the material keeps magnetized after the field is removed |